Topic 4, Part 2
Completely Randomized Design with Two Arms

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Wilcoxon Rank-Sum / Mann-Whitney Test

- A nonparametric alternative to the two independent sample t-test
- Based solely on the order in which the observations from the two samples fall
- H_0: distribution 1 = distribution 2
  - No assumptions about the nature of this distribution (center, spread, skewness, …)
  - Reduces influence of outliers
- For large samples, the test statistic is approximately normal under H_0

Categorical Dependent Variable

Ex): Being Anemic (0,1)

- H_0: P(event|treatment) = P(event|control)
- Chi-square test of independence
- Fisher's exact test (when sample sizes / expected values for specific cells are small)

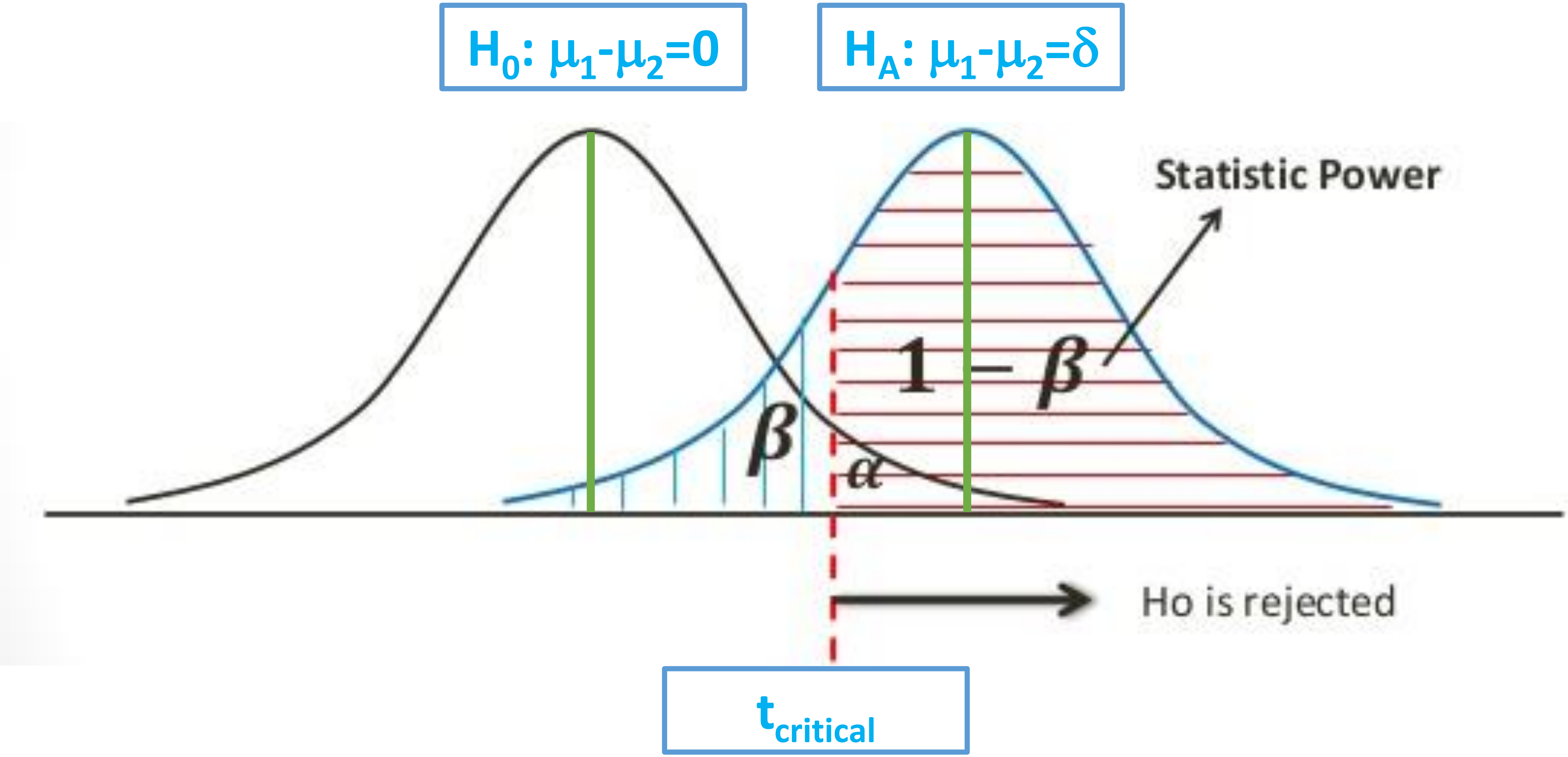
Conclusion | Reality

- P(Type I error) = P(Reject H_0 | H_0 true) = α
- Power = 1-P(Type II error)

  = 1-P(Do not reject H_0 | H_A true)

  = P(Reject H_0 | H_A true)

POSSIBLE CONCLUSIONS		REALITY: H_0 — Treatments are NOT different	REALITY: H_A — Treatments ARE different
Do not reject H_0	We conclude treatments are NOT different from each other	Correct decision (cell a)	Type II error (probability = β) (cell b)
Reject H_0	We conclude treatments ARE different from each other	Type I error (probability = α) (cell c)	Correct decision (cell d)

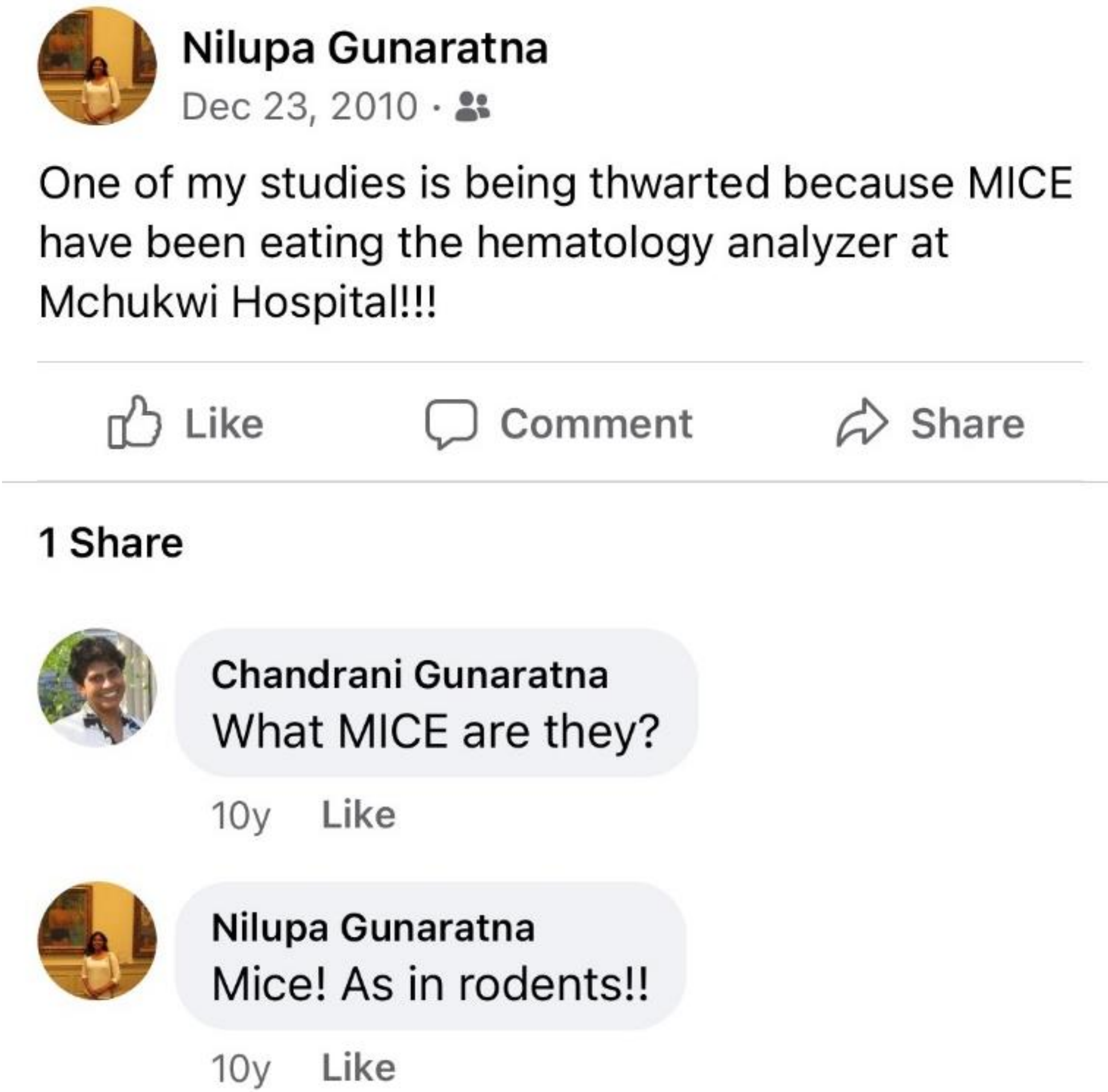


Power increases if:

- Sample size increases
- δ increases
- Variability decreases
- α increases

“power for 2 indep samples.sas”

- Where do all the numbers come from?
- Note none of these relationships are linear
- Note the diminishing returns from increasing sample size
- Actual mean values don’t matter
  - Effect size: difference/SD
- When the calculated n is fractional, round up



- Anticipate loss to follow-up (e.g., increase calculated sample size to allow for 10% loss to follow-up)
- You will also need systems to document (electronically) loss to follow-up (who, when, why)

Paired Comparisons

- Examples:
  - Two potential allergens – both tested on each participant
  - Pre-test vs. post-test
  - This is actually a special case of a randomized complete block design – we'll get to that later in the course
- Control noise – improve precision
- However, observations are no longer independent
- Perform inference on the differences for each pair
- Instead of 2n observations, we have n independent differences
- H_0: average difference=0

Paired Comparisons

- Paired t-test boils down to 1-sample t-test
- Nonparametric option: Wilcoxon signed-rank test
- “paired hb.sas”

Paired Case

- var(aX+bY)

  $=a^2 \text{var}(X) + b^2 \text{var}(Y) + 2ab \text{ cov}(X,Y)$

- $\text{var}(Y_1\text{-}Y_2)$

  $= \text{var}(Y_1) + \text{var}(Y_2) - 2 \text{ cov}(Y_1,Y_2)$

- Pairing is intended to reduce variance. Variance is reduced when there is positive covariance / correlation. However, in the process, you also lose degrees of freedom ½ of sample size
  - Loss of statistical power when covariance low
  - If the pairs are uncorrelated, the difference is twice as noisy